

Faculty of Computer & Information Sciences
Ain Shams University
Course Code: CSC 441
Course Name: Selected Topic (Computer Vision)
Exam: (Final) 26/05/2019
Year: (2nd term) 4th undergraduate (SC)



Instructors: Dr. Dina Khattab
Offering Dept: Scientific Computing
Academic year: 2018-2019
Duration: 3 Hours
Total Grade: 65 Marks
Pages: 2

Question 1: [16 Marks]

- A) Background subtraction takes the (absolute) difference between a background image frame $B(x, y, t)$ and an image frame $I(x, y, t)$, and determines foreground pixels by applying a threshold on the result. [8 Marks]
- Explain two methods where the background image $B(x, y, t)$ can be estimated.
 - What are the disadvantage(s) of each method (if exist)?
- B) In computer vision, a feature is a piece of information which is relevant for solving the computational task related to a certain application. Features may be specific structures in the image such as points, edges or objects. [8 Marks]
- What does it mean if an image feature is invariant to a transformation X ? Give two examples of X .
 - Why is having invariance a good characteristic for an image feature detector?
 - What is the main difference between an image feature detector and feature descriptor? Name an example algorithm for each of them.
 - Explain how you can make your named feature descriptor rotationally invariant.

Question 2: [22 Marks]

- A) Explain briefly the steps of using the **Meanshift** algorithm for object tracking. [7 Marks]
- B) For each of the following terms, **Define** then **Explain** how it may be used to help solve a computer vision problem (and specify the problem) [15 Marks]
- | | | |
|-------------------------|--------------------------------|-------------------|
| i. Second moment matrix | ii. Gaussian pyramid | iii. Bag of Words |
| iv. Kernel trick | v. Dynamics of a moving object | |

Question 3: [15 Marks]

- A) In foreground-background image segmentation, the **Grabcut** is considered one of the fundamental approaches. Answer the following questions about **Grabcut** [8 Marks]
- What image features can be used for segmentation?
 - How initial segmentation of image pixels can be done?
 - What kind of model representation can be formulated to define both foreground and background classes?
 - What is the main technique upon which the Grabcut approach depends for graph partitioning?

- B) Your job is to build a computer vision system to track a news reporter's face based on skin color. Given a video scene with one person in it who is facing the camera – but you don't initially know exactly where – you need to output the (x, y) location of the center of the face in each frame. Describe a complete approach to accomplish this task, including any preparation or training that would be necessary before running the program. Clearly state the program's inputs and outputs. [7 Marks]



Question 4:

[12 Marks]

Given six training points (A, B, C, D, E, F) and five weak-learners (h_1, h_2, h_3, h_4, h_5) which make the following misclassifications

Weak learner	Misclassified Training Points				
h_1	A		D		F
h_2			D		
h_3		B	C		
h_4	A	B			F
h_5		B	C	D	

Perform two iterations of AdaBoost. In each iteration, pick the weak learner with the lowest error rate. Fill in the tables below with your calculations, in addition to the equations of Error rate (err_m), Voting power (α_m) and Weight update (w_i).

Equation of Error rate (err_m):	
Equation of Voting power (α_m):	
Equation of Weight update (w_i):	

	Iteration 1	Iteration 2
Weight _A		
Weight _B		
Weight _C		
Weight _D		
Weight _E		
Weight _F		
Error rate of weak learner		
$h_1 (err_m)$		
$h_2 (err_m)$		
$h_3 (err_m)$		
$h_4 (err_m)$		
$h_5 (err_m)$		
Selected weak learner:		
Voting power (α_m)		

With Best Wishes.

Dr. Dina Khatib